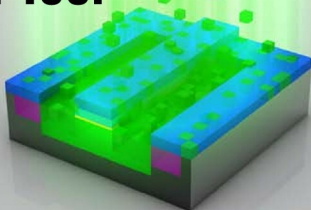


CPUID

The x86 Processor Identification Tool



With CPUID assembly instruction, software can determine information about the processor—the vendor, model, family, supported instruction sets, and other extended features. This article will try to demonstrate the use of CPUID instruction with assembly and C, on Linux.

*T*ry peeking inside the `/proc/cpuid` file:

```
cat /proc/cpuid
```

...and any Linux console will show up details about the processor -- its vendor details, model, family, and capabilities. All this valuable information is available thanks to the CPUID (CPU Identification) instructions set available in x86 processors.

This article is all about getting our hands dirty with CPUID.

CPUID: How things work

CPUID instruction is used to return the capabilities and configuration information

of x86-based processors. But how can we use CPUID to fetch this information?

Well, fundamentally the information is accessed by loading a specific value in the EAX register and executing the CPUID instruction. The CPUID instructions will then perform a specific function based on the input given to EAX. The results are finally returned to registers EAX, EBX, ECX and EDX.

- Input supported value → EAX
- Execute instruction → CPUID
- Result(s) → EAX, EBX, ECX, EDX

But why do we need CPUID? Well any operating system and application software can exploit the CPUID instruction to determine if it is compatible with the